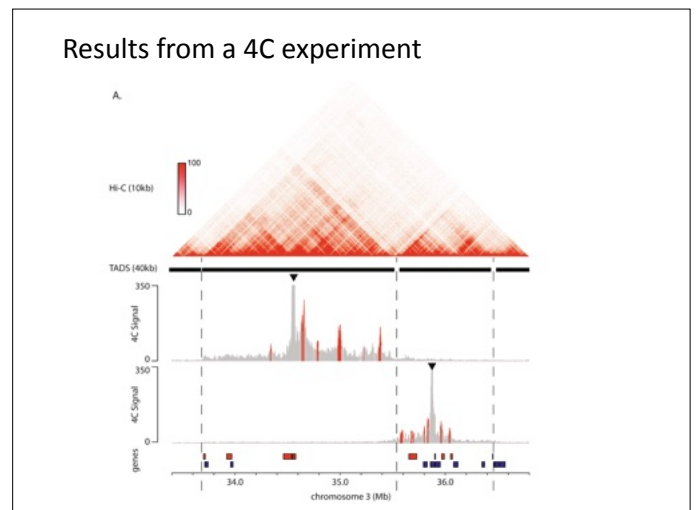
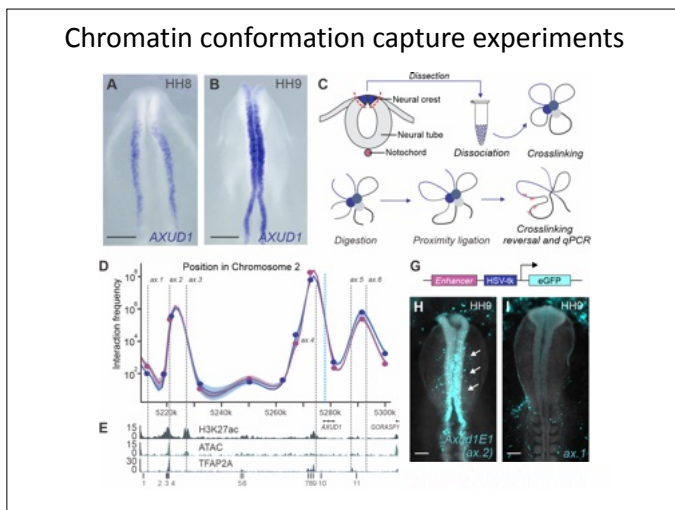
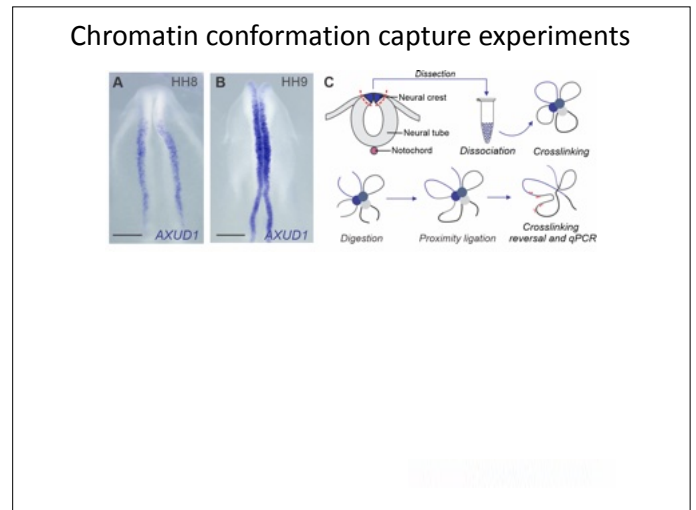
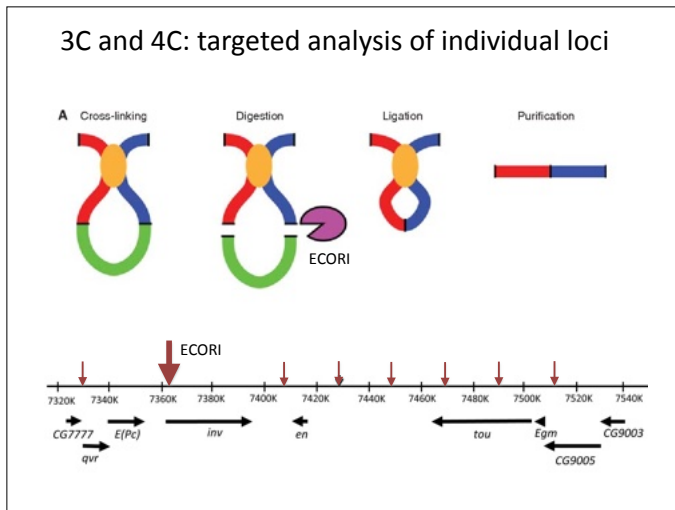
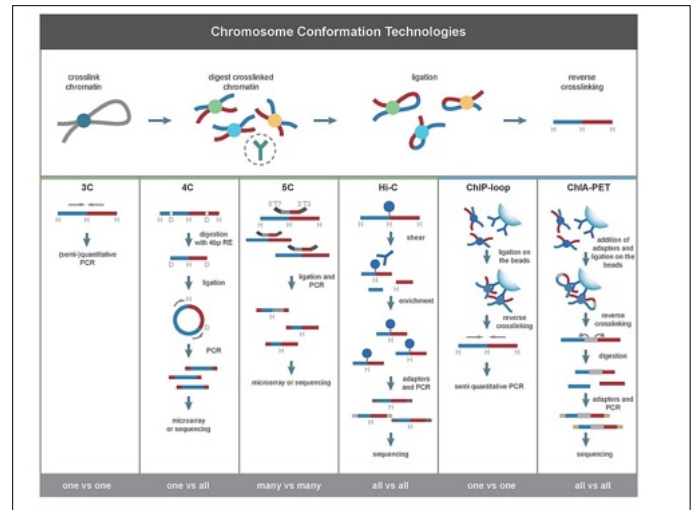
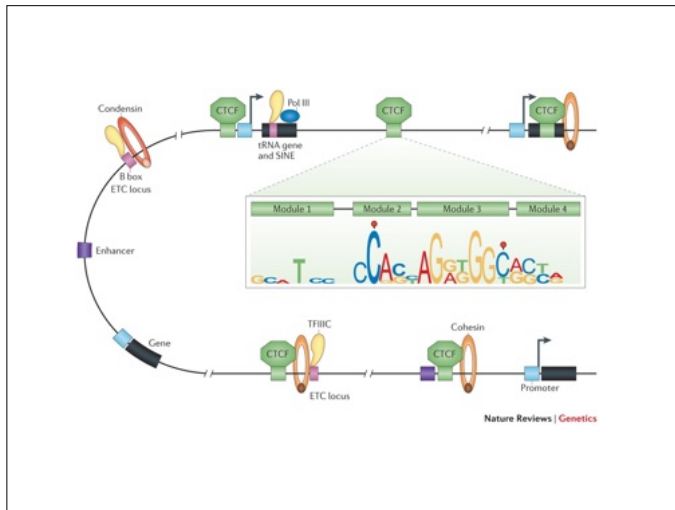
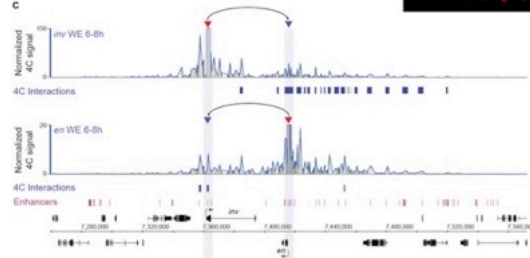
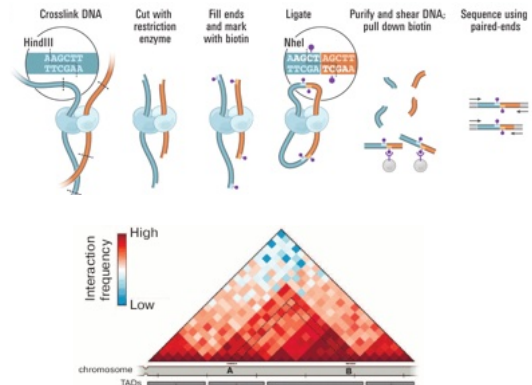




Results from a 4C experiment

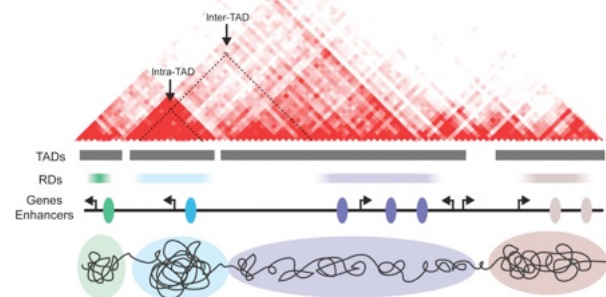


Assembling the genome connectomes with Hi-C

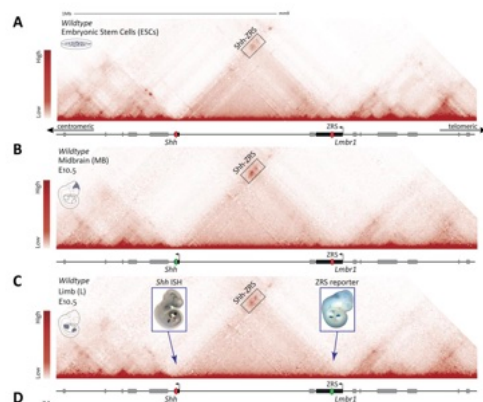


Results from a Hi-C experiment

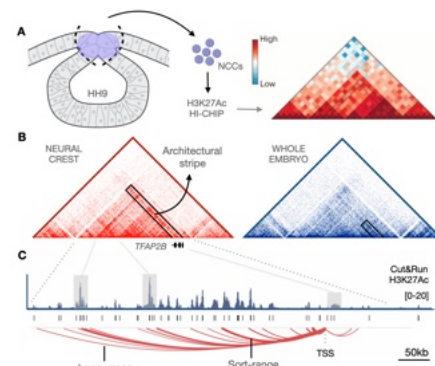
(a)



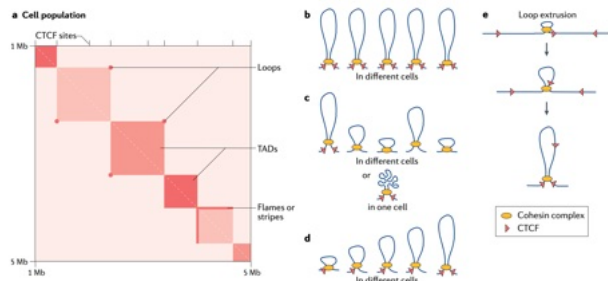
Results from a Hi-C experiment



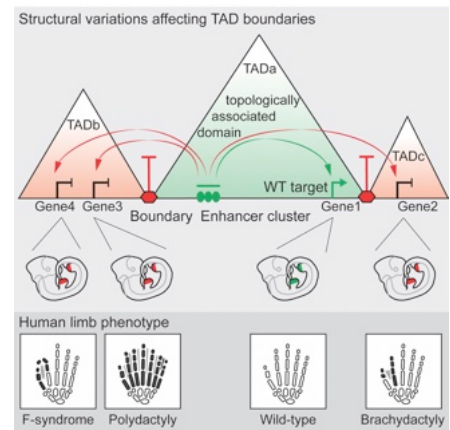
Results from a Hi-C experiment



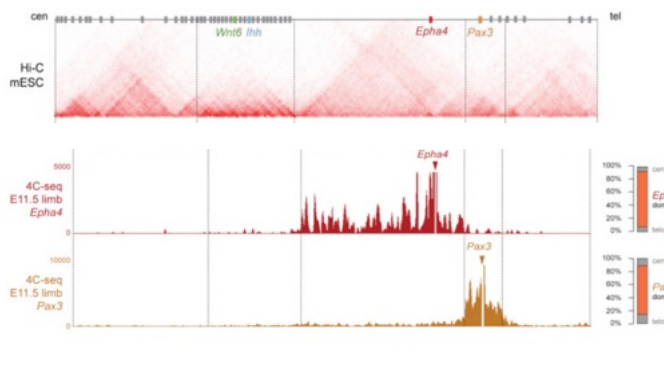
Interpreting Hi-C/Hi-ChIP results:



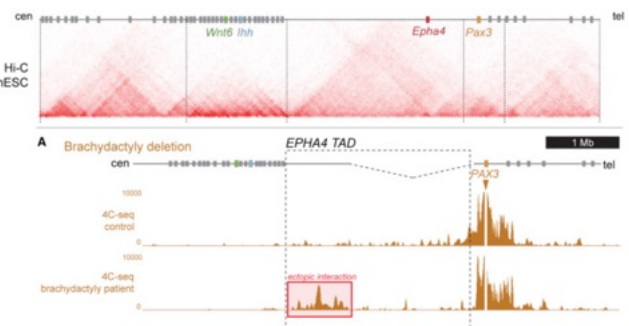
Why is this important?



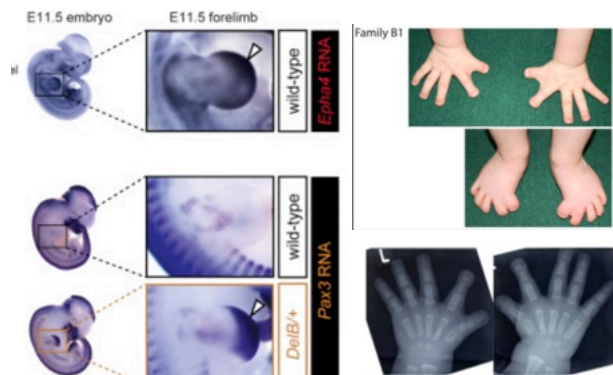
Results from a Hi-C experiment



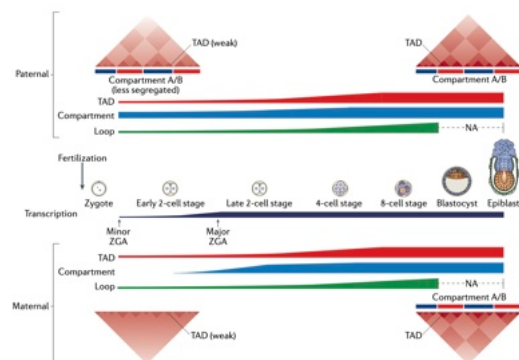
Results from a Hi-C experiment



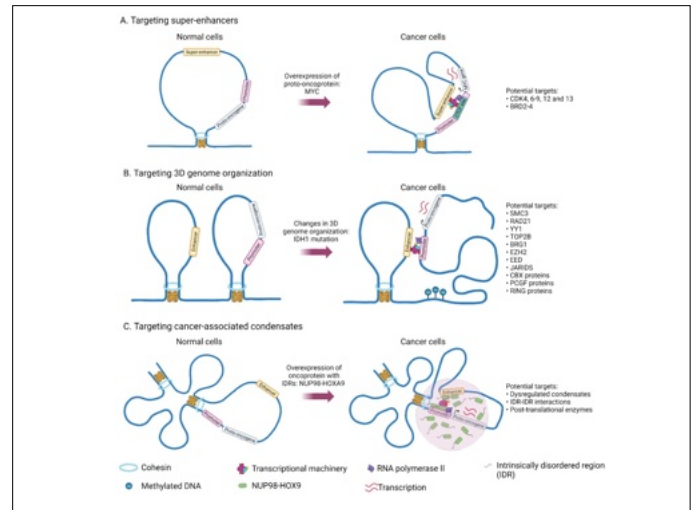
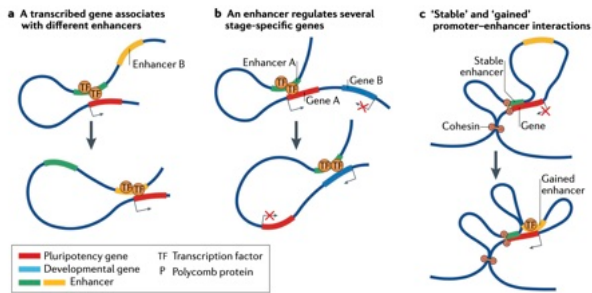
Results from a Hi-C experiment



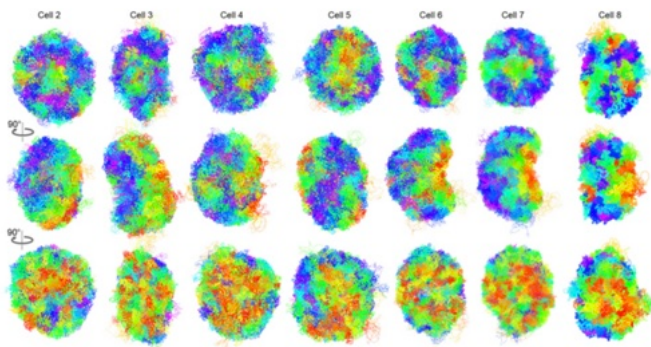
When are compartments formed during development:



Types of intra-tad DNA loops



Single cell Hi-C: is chromatin organization heterogeneous within a cell population?



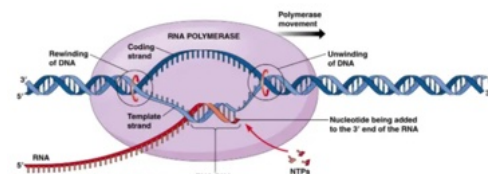
PART IV:

The Transcriptional Machinery

Division of labor in transcription:

Polymerase	Major transcripts	Where	Core biological job
Pol I	45S pre-rRNA → 18S, 5.8S, 28S rRNAs	Nucleus	Build ribosomes
Pol II	pre-mRNA → mRNA also many regulatory RNAs	Nucleoplasm	Regulated gene expression
Pol III	tRNA, 5S rRNA, US snRNA and other small RNAs	Nucleoplasm	Support translation

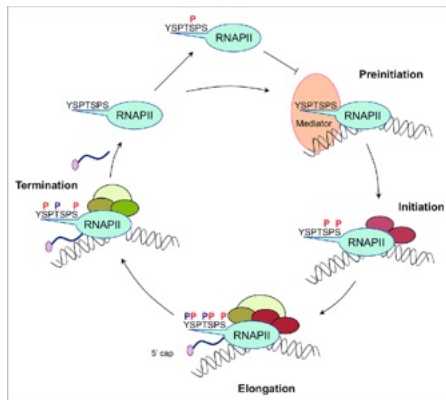
How does Pol II transcribe DNA?



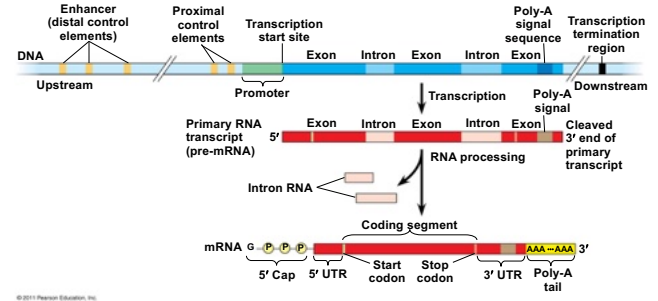
RNA polymerase moves along the DNA template, locally unwinding the duplex and adding complementary nucleotides to the 3' end of the growing RNA.

This creates a short RNA-DNA hybrid inside the polymerase, while DNA rewinds behind the enzyme as transcription proceeds.

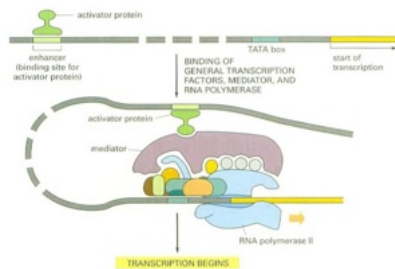
The Pol II transcription cycle:



Organization of a typical eukaryotic gene

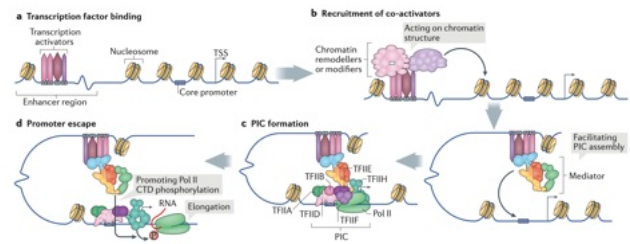


Order of events for initiation of transcription



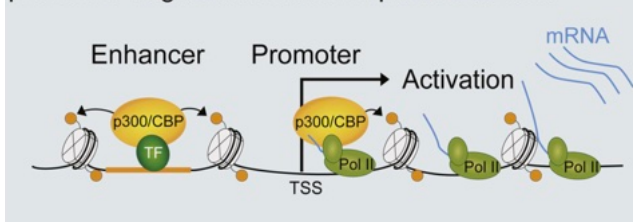
1. TFs bind enhancer
2. Enhancer recruits p300/CBP, Mediator, BAF, cohesin-associated factors
3. Enhancer and promoter come into proximity
4. Promoter-bound factors assemble/stabilize: TFIID, Pol II, GTFs
5. A productive PIC forms
6. Pol II initiates transcription

Order of events for initiation of transcription



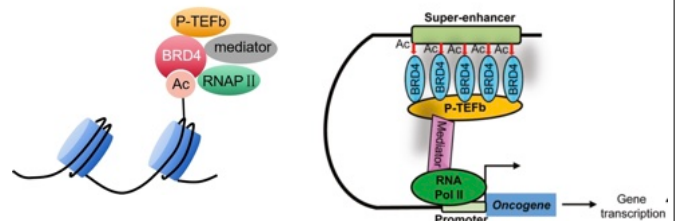
TFs bound to enhancers recruit P300 to active regulatory regions

p300/CBP regulation of transcription activation

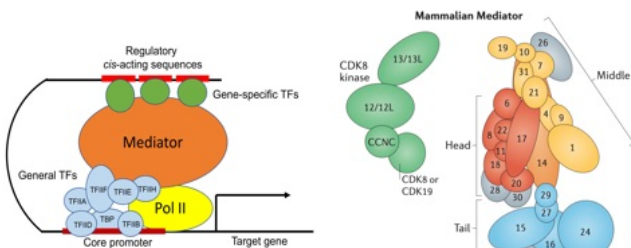


p300/CBP acetylates nearby histones, especially H3K27, creating an active chromatin environment that promotes transcriptional activation.

High levels of histone acetylation (H3K27ac) results in the recruitment of BRD4

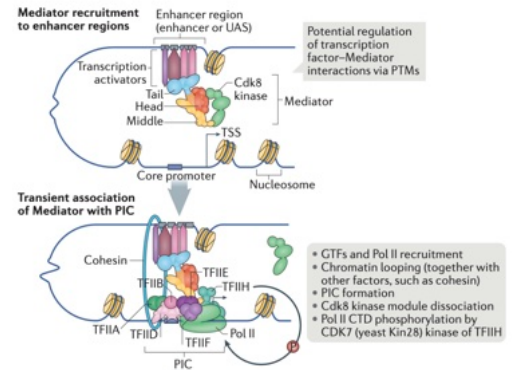


What is the Mediator?

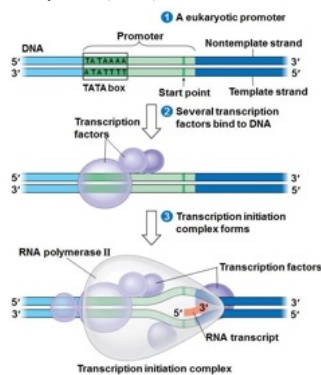


Mediator is a large coactivator complex that connects sequence-specific transcription factors bound at regulatory DNA to the general transcription machinery and RNA polymerase II at the core promoter.

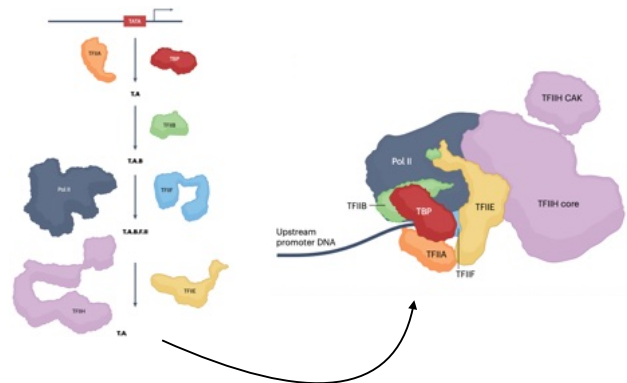
Mediator triggers assembly of the pre-initiation complex (PIC)



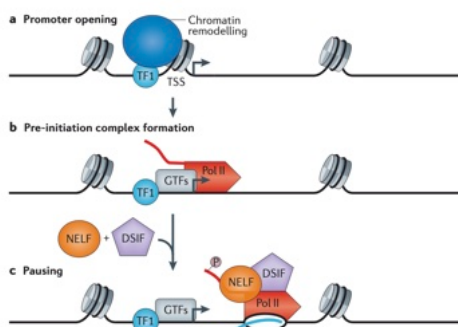
Mediator triggers assembly of the pre-initiation complex (PIC)



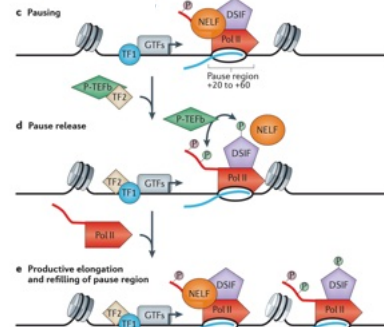
Assembly of the pre-initiation complex



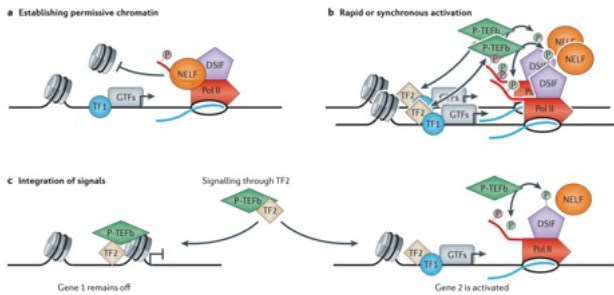
Establishment and release of paused Pol II



Establishment and release of paused Pol II



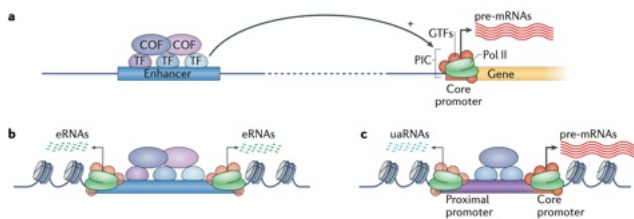
Hypotheses for the functions of Pol II pausing



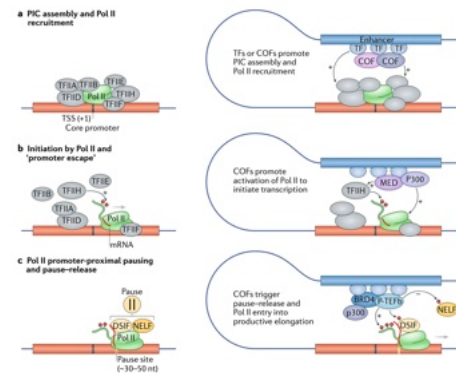
Review of order of events in transcription:

1. Activators (TFs) bind enhancers
2. Coactivators remodel/acetylate chromatin
3. Mediator and GTFs recruit Pol II
4. PIC assembles at the promoter
5. Pol II initiates
6. Pol II pauses
7. P-TEFb releases pausing
8. Productive elongation and RNA processing occur
9. Transcription terminates

Divergent transcription at enhancers and promoters:

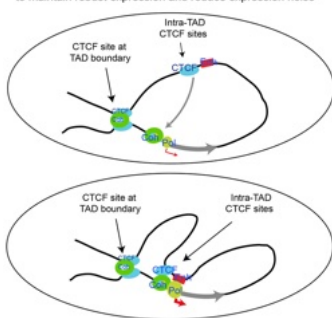


Long-range elements are the control center of transcription:



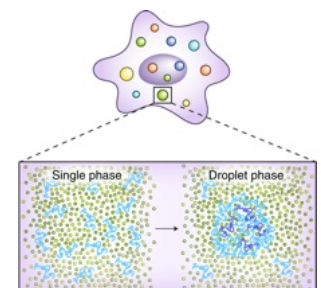
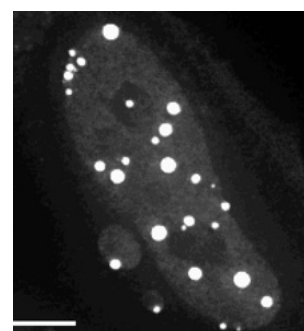
How does binding of TFs trigger DNA loop formation?

Intra-TAD CTCF sites stabilize enhancer-promoter interaction to maintain robust expression and reduce expression noise



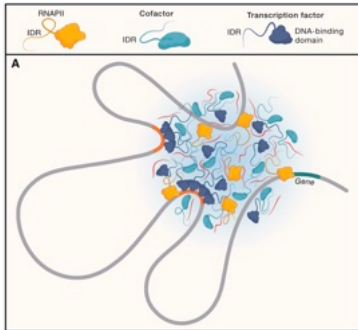
Case 1: recruitment of CTCF to the enhancer

How does binding of TFs trigger DNA loop formation?



Case 2: Liquid phase separation

How does binding of TFs trigger DNA loop formation?



Case 2: Liquid phase separation

Sequential Pol II deployment and transcriptional bursting:

